Lecture 8. Limit Theorems

Markov and Chebyshev, weak and strong laws, central limit theorem, Chernoff bounds, Jensen

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8.1 Markov and Chebyshev inequalities; the weak law

Proposition 8.1 (Markov's inequality)

If $X \geq 0$ and $a > 0$, then $P(X \geq a) \leq E[X]/a$.

Proof. $a\mathbb{1}\{X \geq a\} \leq X$; take expectations. ∎

Proposition 8.2 (Chebyshev's inequality)

If X has mean μ and variance σ^2, then $P(|X - \mu| \geq k) \leq \sigma^2/k^2$ for $k > 0$.

Proof. Apply Markov's inequality to $(X - \mu)^2$ and $a = k^2$. ∎

These bounds use only the mean (and variance), so they are often crude, but they hold for every distribution.

Example 8.1

(a) Weekly output has mean 50. Then $P(X \geq 75) \leq 50/75 = 2/3$. If also $\mathrm{Var}(X) = 25$, $P(40 < X < 60) \geq 1 - 25/100 = 3/4$.
(b) For $X \sim U(0, 10)$, Chebyshev gives $P(|X - 5| > 4) \leq (25/3)/16 \approx 0.52$, whereas the true value is 0.2.
(c) If $\mathrm{Var}(X) = 0$, then $P(|X - \mu| > 1/n) = 0$ for all n, and by continuity $P(X = \mu) = 1$.

Theorem 8.3 (Weak law of large numbers)

If $X_1, X_2, \ldots$ are i.i.d. with mean μ, then for every $\epsilon > 0$, $P(|\bar{X}_n - \mu| \geq \epsilon) \to 0$ as $n \to \infty$.

Proof. Assuming finite variance σ^2: $E[\bar{X}_n] = \mu$, $\mathrm{Var}(\bar{X}_n) = \sigma^2/n$, and Chebyshev gives the bound $\sigma^2/(n\epsilon^2) \to 0$. ∎

8.2 The central limit theorem

Theorem 8.4 (Central limit theorem)

If $X_1, X_2, \ldots$ are i.i.d. with mean μ and variance $\sigma^2 \in (0, \infty)$, then for every a,

$$P\Big(\frac{X_1 + \cdots + X_n - n\mu}{\sigma\sqrt{n}} \leq a\Big) \to \Phi(a) \qquad (n \to \infty).$$

The key tool is: if $M_{Z_n}(t) \to M_Z(t)$ for all t, then $F_{Z_n}(a) \to F_Z(a)$ at continuity points of F_Z (quoted).

Proof sketch. Take $\mu = 0$, $\sigma = 1$, and assume the mgf M of X_i exists near 0. Let $L(t) = \log M(t)$; then $L(0) = 0$, $L'(0) = \mu = 0$, $L''(0) = E[X^2] - \mu^2 = 1$. The mgf of $Z_n = \sum X_i/\sqrt{n}$ is $M(t/\sqrt{n})^n$, and by L'Hôpital twice (in $x = 1/\sqrt{n}$),

$$nL\Big(\frac{t}{\sqrt{n}}\Big) = \frac{L(tx)}{x^2} \to \frac{t^2 L''(0)}{2} = \frac{t^2}{2}.$$

So $M_{Z_n}(t) \to e^{t^2/2}$, the $N(0, 1)$ mgf. (Without an mgf the same argument works with characteristic functions.) ∎

The theorem explains why measurement errors, heights, and sums of many small effects are approximately normal. It also holds for independent, non-identically distributed X_i that are uniformly bounded with $\sum \mathrm{Var}(X_i) = \infty$: then $(\sum(X_i - \mu_i))/\sqrt{\sum \sigma_i^2}$ is approximately $N(0,1)$.

Example 8.2 (How many measurements?)

An astronomer averages n independent measurements with variance 4. To be 95% sure that the average is within 0.5 of the true value, the CLT requires $P(|Z| \le 0.5\sqrt{n}/2) \approx 0.95$, i.e. $0.25\sqrt{n} \ge 1.96$, $n \ge 62$. Chebyshev's inequality would demand $4/(0.25n) \le 0.05$, i.e. $n \ge 320$: guaranteed for every distribution, but far more conservative.

Example 8.3

(a) The sum S of 10 fair dice has mean 35 and variance $350/12$; with the continuity correction, $P(30 \le S \le 40) \approx 2\Phi(5.5/5.40) - 1 \approx 0.69$.
(b) The sum of 10 independent $U(0,1)$ variables has mean 5 and variance $10/12$, so $P(\sum X_i > 6) \approx 1 - \Phi(1.10) \approx 0.137$.
(c) Exam grading times are i.i.d. with mean 20 and standard deviation 4 minutes. The probability that the first 25 exams are graded within 450 minutes is approximately $\Phi((450 - 500)/20) = \Phi(-2.5) \approx 0.006$.

8.3 The strong law of large numbers

Theorem 8.5 (Strong law of large numbers)

If $X_1, X_2, \ldots$ are i.i.d. with finite mean μ, then $P(\lim_{n\to\infty} \bar{X}_n = \mu) = 1$.

Proof. Under the extra assumption $K = E[X_i^4] < \infty$, with $\mu = 0$. Expanding $E[S_n^4]$, the terms with some index appearing exactly once vanish, leaving $nE[X^4] + 6\binom{n}{2}E[X_i^2 X_j^2] \le nK + 3n(n-1)K$ (as $E[X^2]^2 \le E[X^4]$). Hence $E[\sum_n S_n^4/n^4] \le \sum_n (K/n^3 + 3K/n^2) < \infty$, so $\sum_n S_n^4/n^4 < \infty$ with probability 1, which forces $S_n^4/n^4 \to 0$, i.e. $\bar{X}_n \to 0$. ∎

The weak law says that for each large n, $\bar{X}_n$ is *probably* close to μ; the strong law says that, with probability 1, the whole sequence eventually stays close to μ. Applied to $X_i = \mathbb{1}\{E \text{ occurs on trial } i\}$, it justifies the frequency interpretation: the long-run proportion of trials on which E occurs equals $P(E)$ with probability 1.

8.4 Other inequalities

Proposition 8.6 (One-sided Chebyshev)

If $E[X] = 0$ and $\mathrm{Var}(X) = \sigma^2$, then $P(X \ge a) \le \sigma^2/(\sigma^2 + a^2)$ for $a > 0$.

Proof. For $b > 0$, $P(X \ge a) \le P((X+b)^2 \ge (a+b)^2) \le (\sigma^2 + b^2)/(a+b)^2$ by Markov; choose $b = \sigma^2/a$. ∎

Proposition 8.7 (Chernoff bounds)

If $M(t) = E[e^{tX}]$, then $P(X \geq a) \leq e^{-ta}M(t)$ for all $t > 0$ and $P(X \leq a) \leq e^{-ta}M(t)$ for all $t < 0$.

Proof. For $t > 0$, $P(X \geq a) = P(e^{tX} \geq e^{ta}) \leq E[e^{tX}]e^{-ta}$ by Markov. ■

One then minimizes over t.

Example 8.4

(a) Standard normal: $e^{-ta+t^2/2}$ is minimized at $t = a$, so $P(Z \geq a) \leq e^{-a^2/2}$ for $a > 0$.
(b) Poisson(λ) and $i > \lambda$: minimizing $e^{\lambda(e^t-1)-ti}$ at $e^t = i/\lambda$ gives $P(X \geq i) \leq e^{-\lambda}(e\lambda)^i/i^i$.
(c) A gambler wins or loses 1 with probability $\frac{1}{2}$ each play. After n plays, S_n has $M(t) = \cosh(t)^n \leq e^{nt^2/2}$, so $P(S_n \geq a) \leq e^{-a^2/(2n)}$ (take $t = a/n$); e.g. $P(S_{10} \geq 6) \leq e^{-1.8} \approx 0.165$, while the exact value is 0.0547.

Proposition 8.8 (Jensen's inequality)

If f is convex, then $E[f(X)] \geq f(E[X])$ whenever the expectations exist.

Proof. With $\mu = E[X]$, convexity gives $f(x) \geq f(\mu) + f'(\mu)(x - \mu)$ (for twice differentiable f, by Taylor's theorem with $f'' \geq 0$). Take expectations. ■

For example, an investor choosing between a sure return μ and a random return X with mean μ, with concave utility u, prefers the sure amount since $E[u(X)] \leq u(\mu)$.

8.5 The Poisson approximation, quantified

Let $X = \sum_{i=1}^n X_i$ with independent $X_i \sim$ Bernoulli(p_i), and $Y \sim$ Poisson(λ) with $\lambda = \sum p_i$. Coupling each X_i with a Poisson(p_i) variable Y_i so that they agree with probability at least $1 - p_i^2$, one obtains, for every set A of integers,

$$|P(X \in A) - P(Y \in A)| \leq \sum_{i=1}^{n} p_i^2.$$

For Bin(n, p) the bound is $np^2 = \lambda p$: the approximation is good when p is small, whatever n is.

Example 8.5 (Birthdays, again)

Let $X_1, X_2, \ldots$ be the birthdays of people arriving in sequence and N the number needed for the first shared birthday. For n people, the event $N > n$ means that none of the trials "person j shares with one of persons $1, \ldots, j-1$" succeeds; these have success probabilities $(j-1)/365$ given no earlier success, and an exact product gives $P(N > n) = \prod_{j=1}^n (1 - \frac{j-1}{365}) \approx \exp\{-\sum_j \frac{j-1}{365}\} = e^{-n(n-1)/730}$, the Poisson-paradigm answer of Example 4.6.